

Maximum Likelihood Estimation (MLE)

Subjective Problems

Problem 1

In random sampling from normal population $N(\mu; \sigma^2)$, find the maximum likelihood estimators for:

- (i) μ when σ^2 is known.
- (ii) σ^2 when μ is known.
- (iii) the simultaneous estimation of μ and σ^2 .

Problem 2

Find the maximum likelihood estimate for the parameter λ of a Poisson distribution on the basis of a sample of size n . Also find its variance.

Problem 3

It is known that a sample consisting of the values 12, 11.2, 13.5, 12.3, 13.8, and 11.9 comes from a population with the density function

$$f(x; \theta) = \begin{cases} \frac{\theta}{x^{(\theta+1)}} & \text{if } x > 1, \\ 0 & \text{elsewhere} \end{cases},$$

where $\theta > 0$. Find the maximum likelihood estimate of θ .

Problem 4

Prove that the maximum likelihood estimate of the parameter α of a population having density function: $\frac{2}{\alpha^2}(\alpha - x)$, $0 < x < \alpha$, for a sample of unit size is $2x$, x being the sample value. Show also that the estimate is biased.

Problem 5

Find the maximum likelihood estimator of parameter θ of a population having density function $(1 + \theta)x^\theta$.

Problem 6

Suppose 10 rats are used in a biomedical study where they are injected with cancer cells and then given a cancer drug that is designed to increase their survival rate. The survival times, in months, are 14, 17, 27, 18, 12, 8, 22, 13, 19, and 12. Assume that the exponential distribution applies. Give a maximum likelihood estimate of the mean survival time.

Problem 7

A random sample of size $n = 5$ is drawn from a normal distribution with unknown mean μ and known variance $\sigma^2 = 4$. The observed sample values are:

2, 4, 5, 3, 6

- (a) Find the Maximum Likelihood Estimator (MLE) of μ .
- (b) Calculate the estimated value of μ using the given data.

Problem 8

Prove that the MLE of variance for a normal distribution is:

$$\hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$$

Problem 9

A random sample x_1, x_2, x_3, x_4 is drawn from an exponential distribution:

$$f(x; \lambda) = \lambda e^{-\lambda x}, \quad x > 0$$

Given data: 2, 1, 3, 2

Find the MLE of λ and compute its value.

Problem 10

Suppose that X is a discrete random variable with the following probability mass function, where $0 \leq \theta \leq 1$ is a parameter.

X	0	1	2	3
$f(x)$	$2\theta/3$	$\theta/3$	$2(1 - \theta)/3$	$(1 - \theta)/3$

The following 10 independent observations were taken from such a distribution: (3, 0, 2, 1, 3, 2, 1, 0, 2, 1). What is maximum likelihood estimate of θ .

MCQs Problems

Q10. A Poisson sample gives mean = 6. Find MLE of λ .

- A. 4
- B. 5
- C. 6
- D. 7

Answer: C. 6

Q11. A Bernoulli sample has 8 successes in 20 trials. Find MLE of p .

- A. 0.2
- B. 0.3
- C. 0.4

D. 0.5

Answer: C. 0.4

Q12. In a Binomial distribution, $x=30$ successes out of $n=50$. Find MLE of p .

A. 0.4

B. 0.5

C. 0.6

D. 0.7

Answer: C. 0.6

Q13. For an exponential distribution, sample mean = 4. Find MLE of λ .

A. 0.20

B. 0.25

C. 0.50

D. 1.00

Answer: B. 0.25

Q14. For a Normal sample with mean 12, MLE of μ is:

A. 10

B. 11

C. 12

D. 13

Answer: C. 12

Q15. For sample 3, 5, 7, 9, find MLE of μ .

A. 5

B. 6

C. 7

D. 8

Answer: B. 6

Q16. If $x=15$ and $n=25$ in a Binomial model, find MLE of p .

A. 0.4

B. 0.5

C. 0.6

D. 0.7

Answer: C. 0.6

Q17. For Normal distribution, MLE of variance is:

A. $\Sigma(x-\bar{x})^2/n$

B. $\Sigma(x-\bar{x})^2/(n-1)$

C. $\Sigma x^2/n$

D. $\bar{x}^2$

Answer: A. $\Sigma(x-\bar{x})^2/n$

Q18. For Bernoulli sample with 12 successes in 30 trials, find $\hat{p}$.

A. 0.3

B. 0.4

C. 0.5

D. 0.6

Answer: B. 0.4

Q19. For Poisson distribution, total count =100 from 20 observations. Find $\hat{\lambda}$.

A. 4

B. 5

C. 6

D. 7

Answer: B. 5

Q20. Exponential sample mean =2. Find $\hat{\lambda}$.

A. 0.25

B. 0.50

C. 0.75

D. 1.00

Answer: B. 0.50

Q21. In Binomial distribution, $x=45$ and $n=60$. Find MLE of p .

A. 0.60

B. 0.65

C. 0.70

D. 0.75

Answer: D. 0.75

Q.22.The maximum likelihood estimator of parameter θ of a parameter having density function

$(1 + \theta)x^\theta$ is-

$$(a) \text{Log}L(\theta) = -1 - \frac{n}{\sum_{i=1}^n \log x_i}$$

$$(b) \text{Log}L(\theta) = n \log(1 + \theta) + \theta \sum_{i=1}^n \log x_i$$

$$(c) \text{Log}L(\theta) = n \log(1 + \theta)$$

$$(d) \text{Log}L(\theta) = n \log(1 + \theta)$$

Ans B

Q23.Variance of M.L.E is given by-

$$(a) V(\hat{\theta}) = \frac{1}{E(-\frac{\partial^2}{\partial \theta^2} \log L)}$$

$$(b) V(\hat{\theta}) = \frac{2}{E(-\frac{\partial^2}{\partial \theta^2} \log L)}$$

$$(c) V(\hat{\theta}) = \frac{1}{E(\frac{\partial^2}{\partial \theta^2} \log L)}$$

$$(d) V(\hat{\theta}) = \frac{1}{5E(-\frac{\partial^2}{\partial \theta^2} \log L)}$$

Ans A